RL2 - Exercises and solution sketches

NOTE: For some of the exercises the solution is just sketched in a brief manner. Note that a proper solution would require more detailed descriptions, explanations, and in some cases examples. Some of the exercises may have more than one solution, and we just show one of them.

Selected official solutions to exercises from HMU can also be found here: `http://infolab.stanford.edu/~ullman/ialcsols/sols.html`.

Exercise RL2.1. Building and understanding NFAs Solve exercise HMU 2.3.4 (copied below for your convenience).

! **Exercise 2.3.4:** Give nondeterministic finite automata to accept the following languages. Try to take advantage of nondeterminism as much as possible.

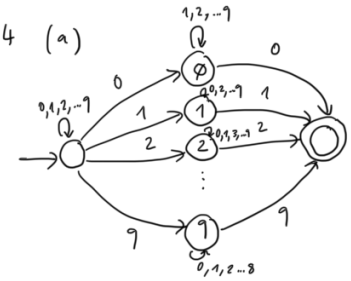
* a) The set of strings over alphabet $\{0, 1, \ldots, 9\}$ such that the final digit has appeared before.

b) The set of strings over alphabet $\{0, 1, \ldots, 9\}$ such that the final digit has *not* appeared before.

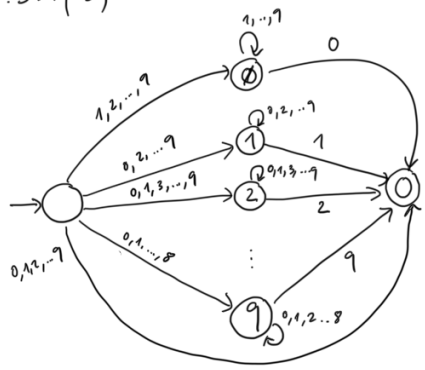
c) The set of strings of 0's and 1's such that there are two 0's separated by a number of positions that is a multiple of 4. Note that 0 is an allowable multiple of 4.

Exercise RL2.1. Building and understanding NFAs - solution

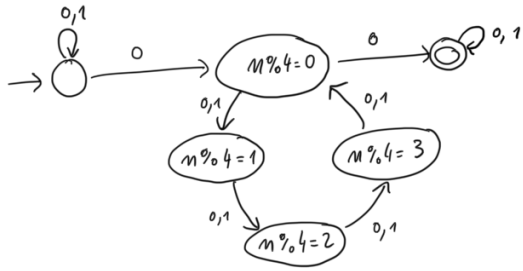
2.3.4 (a)



2.3.4 (b)



2.3.4 (c)



In (a) and (b) the name of the middle states (0,1,..9) indicates the "guess" of the digit. In (c) the name of the states in the loop indicate how many symbols between the two 0s we have observed (modulo 4).

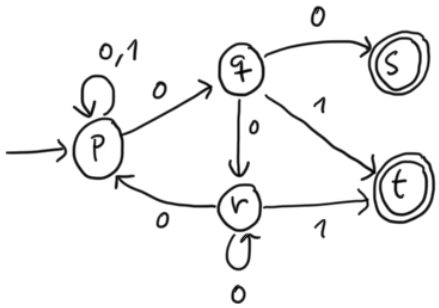
Exercise RL2.2. NFAs to DFAs Solve exercise HMU 2.3.3 (copied below for your convenience).

! Exercise 2.3.3: Convert the following NFA to a DFA and informally describe the language it accepts.

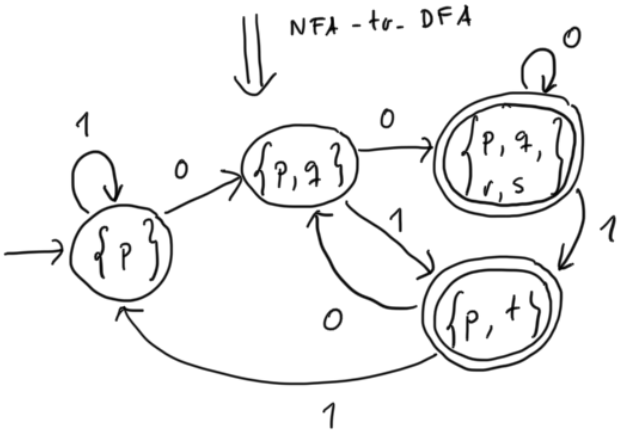
	0	1
$\rightarrow p$	$\{p, q\}$	$\{p\}$
q	$\{r, s\}$	$\{t\}$
r	$\{p, r\}$	$\{t\}$
$*s$	$\emptyset$	$\emptyset$
$*t$	$\emptyset$	$\emptyset$

Exercise RL2.2. NFAs to DFAs - solution

Exercise 2.3.3



NFA - to - DFA



Strings of 0's and 1's that finish with

00 or 01

Exercise RL2.3. On ϵ-NFAs Solve exercises HMU 2.5.1. In 2.5.1(b) consider strings of length 2 or less (instead 3 or less as the book says).

*** Exercise 2.5.1:** Consider the following ϵ-NFA.

	ϵ	a	b	c
$\rightarrow p$	$\emptyset$	$\{p\}$	$\{q\}$	$\{r\}$
q	$\{p\}$	$\{q\}$	$\{r\}$	$\emptyset$
$*r$	$\{q\}$	$\{r\}$	$\emptyset$	$\{p\}$

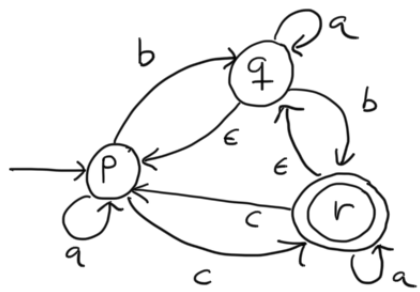
a) Compute the ϵ-closure of each state.

b) Give all the strings of length three or less accepted by the automaton.

c) Convert the automaton to a DFA.

Exercise RL2.3. On ϵ-NFAs - solution

Exercise 2.5.1



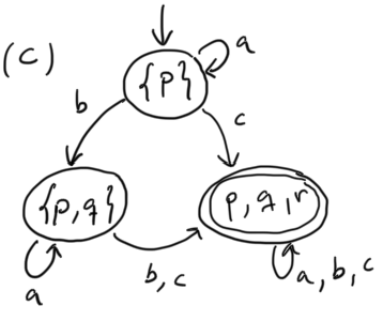
(a)

	ECLOSE
p	{p}
q	{q, p}
r	{r, q, p}

(b)

w	accept?
ε	✗
a	✗
b	✗
c	✓
a a	✗
a b	✗
a c	✓
b a	✗
b b	✓
b c	✓
c a	✓
c b	✓
c c	✓

(c)



Exercise RL2.4. On ϵ-NFAs Solve exercises HMU 2.5.3(b).

Exercise 2.5.3: Design ϵ-NFA's for the following languages. Try to use ϵ-transitions to simplify your design.

a) The set of strings consisting of zero or more a's followed by zero or more b's, followed by zero or more c's.

! b) The set of strings that consist of either 01 repeated one or more times or 010 repeated one or more times.

! c) The set of strings of 0's and 1's such that at least one of the last ten positions is a 1.

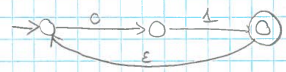
Exercise RL2.4. On ϵ-NFAs - solution

02141 /HRW

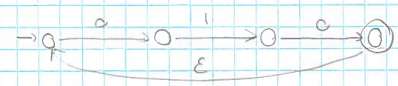
Exercise HMU 2.5.3 (b)

Design an ε-NFA for the set of strings consisting of either 01 repeated one or more times or 010 repeated one or more times.

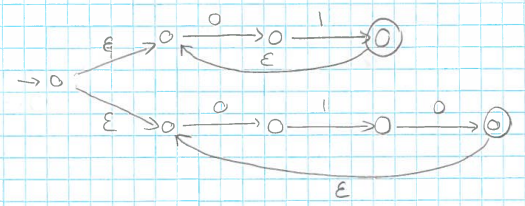
The following ε-NFA will accept strings with 01 repeated one or more times:



The following ε-NFA will accept strings with 010 repeated one or more times:



For the combined language we therefore take



Exercise RL2.5. Correctness of the subset construction In class we have seen the proof of correctness of the subset construction, i.e. that the language of an NFA $N = (Q, \Sigma, \delta_N, q_0, F)$ is the same as the language of the DFA $D = (Q_D, \Sigma, \delta_D, \{q_0\}, F_D)$ obtained by the subset construction. The key property prove is $\forall q \in Q, w \in \Sigma^*.\delta_N^*(q, w) = \delta_D^*(\{q\}, w)$. Prove this again on your own. HINT: Use induction on the length of w or alternatively on the structure of w.

Exercise RL2.5. Correctness of the subset construction - solution The correctness of the subset construction is formally stated in Theorem 2.11 of HMU. The book provides a proof (page 63).

Hungry of more exercises? Check the HMU book (exercises 2.2.X). Recall that Selected official solutions to exercises from HMU can also be found here: `http://infolab.stanford.edu/~ullman/ialcsols/sols.html`.